

Forecast a VAR Model

This example shows how to use `vgxpred` to forecast a VAR model.

[Open Script](#)

`vgxpred` enables you to generate forecasts with error estimates. `vgxpred` requires:

- A fully-specified model (for example, `impSpec` in what follows)
- The number of periods for the forecast (for example, `FT` in what follows)

`vgxpred` optionally takes:

- An exogenous data series
- A presample time series (e.g., `Y(end-3:end,:)` in what follows)
- Extra paths

Load the `Data_USEconModel` data set. This example uses two time series: the logarithm of real GDP, and the real 3-month T-bill rate, both differenced to be approximately stationary. Suppose that a VAR(4) model is appropriate to describe the time series.

```
load Data_USEconModel
DEF = log(DataTable.CPIAUCSL);
GDP = log(DataTable.GDP);
rGDP = diff(GDP - DEF); % Real GDP is GDP - deflation
TB3 = 0.01*DataTable.TB3MS;
dDEF = 4*diff(DEF); % Scaling
rTB3 = TB3(2:end) - dDEF; % Real interest is deflated
Y = [rGDP,rTB3];
```

Fit a VAR(4) model specification:

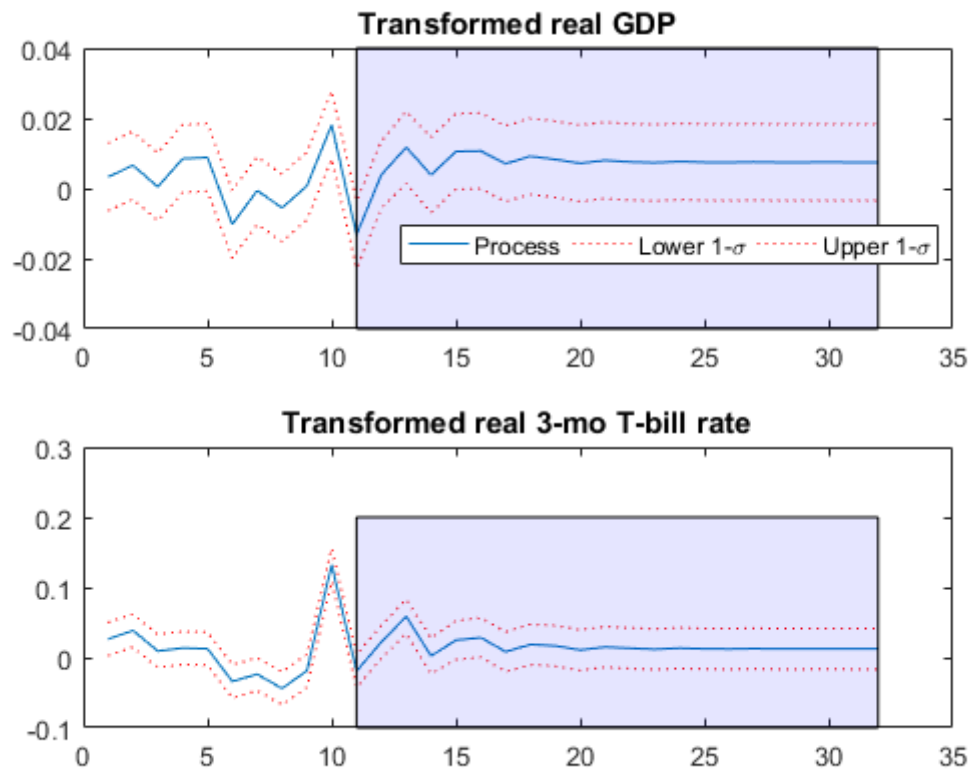
```
Spec = vgxset('n',2,'nAR',4,'Constant',true);
impSpec = vgxvarx(Spec,Y(5:end,:),[],Y(1:4,:));
impSpec = vgxset(impSpec,'Series',...
    {'Transformed real GDP','Transformed real 3-mo T-bill rate'});
```

Predict the evolution of the time series:

```
FDates = datenum({'30-Jun-2009'; '30-Sep-2009'; '31-Dec-2009';
    '31-Mar-2010'; '30-Jun-2010'; '30-Sep-2010'; '31-Dec-2010';
    '31-Mar-2011'; '30-Jun-2011'; '30-Sep-2011'; '31-Dec-2011';
    '31-Mar-2012'; '30-Jun-2012'; '30-Sep-2012'; '31-Dec-2012';
    '31-Mar-2013'; '30-Jun-2013'; '30-Sep-2013'; '31-Dec-2013';
    '31-Mar-2014'; '30-Jun-2014' });
FT = numel(FDates);
[Forecast,ForecastCov] = vgxpred(impSpec,FT,[],...
    Y(end-3:end,:));
```

View the forecast using `vgxplot`:

```
vgxplot(impSpec,Y(end-10:end,:),Forecast,ForecastCov);
```



See Also

[vgxpred](#) | [vgxproc](#) | [vgxvarx](#)

Related Examples

- [Fit a VAR Model](#)
- [Forecast a VAR Model Using Monte Carlo Simulation](#)
- [VAR Model Case Study](#)

More About

- [Vector Autoregressive \(VAR\) Models](#)
- [VAR Model Forecasting, Simulation, and Analysis](#)